

Yumeng He

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RESEARCH INTERESTS

My research interests are computer vision, graphics, and robotics, with a focus on physical AI, object/scene generation, articulated and kinematic asset modeling, physics-based simulation, real-to-sim pipelines, and policy learning.

EDUCATION

University of California, Los Angeles

PhD of Computer Science

- Computer Science major
- Advisor: Prof. Chenfanfu Jiang

Sep 2026 – May 2030

Los Angeles, CA, USA

University of Southern California

Master of Science with Thesis

- Cumulative GPA: 3.91/4.0
- Computer Science major
- Advisor: Prof. Jernej Barbič

Aug 2024 – May 2026

Los Angeles, CA, USA

University of Toronto St. George Campus

Honours Bachelor of Science with High Distinction

- Cumulative GPA: 3.71/4.0
- Computer Science and Mathematics double major

Sep 2019 – May 2024

Toronto, ON, Canada

PUBLICATIONS AND MANUSCRIPTS

- [1] **Fishbone: From One 3D Asset to a Million Controllable Edits**
Yumeng He, Xiaoying Wang, Peihao Li, Yanjia Huang, Joe Masterjohn, Jiajun Wu, Leonidas Guibas, Yin Yang, Ying Jiang, Chenfanfu Jiang
arXiv, 2026
- [2] **SeeClear: Reliable Transparent Object Depth Estimation via Generative Opacification**
Xiaoying Wang*, **Yumeng He***, Jingkai Shi*, Jiayin Lu, Yin Yang, Ying Jiang, Chenfanfu Jiang
European Conference on Computer Vision (ECCV), 2026
- [3] **VoroLight: Learning Quality Volumetric Voronoi Meshes from General Inputs**
Jiayin Lu*, Ying Jiang*, **Yumeng He***, Yin Yang, Chenfanfu Jiang
arXiv, 2025
- [4] **SPARK: Sim-ready Part-level Articulated Reconstruction with VLM Knowledge**
Yumeng He*, Ying Jiang*, Jiayin Lu*, Yin Yang, Chenfanfu Jiang
Computer Vision and Pattern Recognition (CVPR), 2026 **Oral (top 3.45%)**
- [5] **Birth of a Painting: Differentiable Brushstroke Reconstruction**
Ying Jiang*, Jiayin Lu*, Yunuo Chen*, **Yumeng He**, Kui Wu, Yin Yang, Chenfanfu Jiang
Conditionally Accepted by ACM Transactions on Graphics (SIGGRAPH), 2026
- [6] **Self-supervised dual-layer 2D normalizing flow method for industrial anomaly detection**
Zhenlian Miao, Guangzhu Chen, Xiaojuan Liao, Jiu Dai, **Yumeng He**
Applied Soft Computing (ASOC), 2024.

RESEARCH EXPERIENCE

USC Graphics | Master Thesis

Supervisor: Prof. Jernej Barbič

- Enhancing traditional Kirchhoff-Love shell models by introducing a novel elastic energy formulation that accurately captures stretching, compression, and bending, while overcoming KL shells' inability to reproduce Poisson effects

Aug 2025 – present

Los Angeles, CA, USA

UCLA AIVC Lab | Visiting student

Supervisor: Prof. Chenfanfu Jiang

- Leading research on end-to-end articulation-aware 3D mesh generation, designing a pipeline that synthesizes multi-part objects from text or image conditions while mitigating over-segmentation via per-part image-guided local attention
- Performing joint optimization post-generation to infer plausible URDFs, enabling deployment in embodied simulation environments and supporting per-part texturing for visual realism and downstream manipulation compatibility

Jun 2025 – present

Los Angeles, CA, USA

USC RESL Lab | Research Assistant
Supervisor: Prof. Gaurav S. Sukhatme

Jul 2025 – present
Los Angeles, CA, USA

- Designed a few-shot robotic manipulation task that uses Model Predictive Control to optimize material parameters for physically accurate simulation
- Led the Real2Sim pipeline for this project, building a digital twin from a single RGB D image of a cluttered tabletop scene with object segmentation, mesh reconstruction, and pose, scale, and placement estimation, and importing the reconstructed scene into ManiSkill3 for simulation

CDUT | Research Assistant
Supervisor: Prof. Guangzhu Chen

May 2023 - Aug 2023
Remote

- Faced with unlabeled industrial defects on MVTec AD and tasked to raise unsupervised detection/localization accuracy, proposed SS-DualFlow: a dual-layer 2D normalizing-flow that maps features to a Gaussian base to curb information loss and inserts an Exponential Space Attention module to focus on anomaly-salient regions. Result: image-level AUROC = 99.38% and pixel-level AUROC = 98.38% on MVTec AD; co-authored the Applied Soft Computing (2024) paper.

RELEVANT PROJECTS

Incompressible Fluid Simulation: A Comparison | C++, OpenGL

Spring 2025

- Implemented and benchmarked four 2D incompressible fluid solvers, including Stable Fluids, Smoothed Particle Hydrodynamics (SPH), Particle In Cell (PIC), and Affine Particle In Cell (APIC) on identical scenarios, reporting performance-vs-accuracy trade-offs.

Collision Detection with Penalty Method and IPC | C++

Spring 2025

- Built a physically accurate 3D jello cube simulator using a mass-spring system with structural, shear, and bend springs to model real-world deformation under force
- Implemented penalty-based collision detection to handle interactions with static obstacles, including inclined planes and spheres, ensuring realistic response under contact

Inverse Kinematics with Skinning | C++

Spring 2025

- Implemented Linear Blend Skinning and Dual Quaternion Skinning for 3D character deformation with smooth, realistic joint articulation
- Built an end-to-end FK–IK pipeline, supporting forward kinematics for pose propagation and inverse kinematics under joint constraints

Motion Capture Interpolation | C++

Spring 2025

- Implemented four interpolation schemes (including linear & Bézier) with both Euler- and quaternion-based rotations to reconstruct and smooth optical mocap, mitigating gimbal lock and interpolation artifacts

INDUSTRY EXPERIENCE

Software Developer Intern | Full Time
HCL Canada Inc.

Aug 2022 – Aug 2023
Toronto, ON, Canada

COMMUNITY SERVICES

- Member of Society of Women Engineers (SWE)2025 - 2026
- Mentor for Viterbi Graduate Mentorship ProgramFall 2025
- Mentor for Women in Engineering MentorshipFall 2025
- Mentor for Viterbi Graduate Mentorship ProgramSummer 2025

INVITED TALKS

- MPC Is All You Need. [Co-presented]. USC SLURM Lab, hosted by Prof. Daniel SeitaAug 2025

AWARDS & HONORS

- Dean’s List Scholar2020 - 2024
- Ranked in the top 25% of contestants in the Galois Contest2017
- Ranked in the top 25% of contestants in the Cayley Contest2017
- Ranked in the top 25% of contestants in the Canadian Intermediate Mathematics2016

TECHNICAL SKILLS

- Programming Tools:** Linux, Windows, MacOS, C++, Python, HTML, CSS, JavaScript, MySQL, R, OpenGL, GLSL, Eigen, Docker, Jenkins, Gradle, Ant, Shell, Git, Github Actions